
Autex: A Reliability-First In-Car Voice Assistant Agent for the CAR-bench IJCAI-ECAI 2026 Challenge

Nahed Innovation · CAR-bench Track 2 · IJCAI-ECAI 2026

ABSTRACT

We present Autex, a reliability-first in-car voice assistant agent for the CAR-bench Challenge. Autex combines a deterministic multi-layer reliability kernel with a configurable LLM verifier to address three core CAR-bench failure modes: hallucinating unavailable capabilities, acting on ambiguous intent, and violating domain policies. Epistemic guards fire before any LLM call, reducing the Pass@3 to Pass^3 consistency gap. We validate on 369 official trials across four frontier models, identify hallucination_missing_tool_response as the dominant systemic weakness (0–14.3% Pass^3 across all models), and describe our design choices to address it. All model and provider configuration is environment-variable-driven for Track 2 auditability.

1 INTRODUCTION

CAR-bench [Kirmayr et al., 2026] evaluates multi-turn, tool-using LLM agents in an automotive in-car voice assistant domain under uncertainty, ambiguity, and capability constraints. Unlike benchmarks that measure task success under idealized conditions, CAR-bench scores *epistemic reliability*: whether agents know when to act, when to gather more information, and when to refuse. The primary metric is Pass^k, requiring all k independent trials to succeed — penalising inconsistency heavily.

Our system, Autex, is built around a key observation from validation data: the largest driver of low Pass^3 is not a failure to *know* the right answer, but inconsistent *application* of that knowledge. Pass@3 ran 31.7 pp above Pass^3 for Claude Sonnet 4.6 across 369 official trials. Closing this gap is the primary design goal.

2 SYSTEM ARCHITECTURE

2.1 Reliability Kernel

The reliability kernel is a deterministic, zero-LLM-call layer that fires before any model invocation. Five hard rules enforced unconditionally:

- **Tool availability guard:** absent required tool → refuse_or_defer immediately.
- **Parameter support guard:** required parameter absent from tool schema → refuse rather than call with invalid argument.
- **Result completeness guard:** observed tool result null or missing expected field → refuse rather than infer vehicle state.
- **Domain policy guards:** weather checked before sunroof; sunshade fully open before sunroof; rain requires user confirmation.
- **Disambiguation policy:** resolve from preferences / context first; query user only when unresolvable internally.

These guards implement the key insight from our ablation: the missing_tool_response subtype scored 0–14.3% Pass^3 across all four frontier models, while missing_parameter reached 75–100% Pass^3. Parameter availability is checkable at schema-inspection time; result completeness requires runtime verification — so the result completeness guard extends deterministic checking to the execution layer.

2.2 MPAE: Morphological Predictive Allocation Engine

Before any LLM call, Autex runs a deterministic vehicle health scorer over current telemetry. Three components:

- **Bat Future Echo:** projects forward risk across vibration, temperature, LTFT drift, health score, and RPM deviation.

- **Hermit Stability Test:** scores long-term recommendation stability over quick-fix optimisation.
- **Squid Dynamic Objective:** reweights safety / uptime / cost / comfort by inferred context (normal / critical / fleet / budget).

MPAE outputs a maintenance priority (0–100) and strategy: monitor, stabilize, service-soon, or stop-and-inspect. A critical-context override escalates immediately when healthScore < 60, vibration > 95, temp > 105°C, or [LTFT] > 15 — regardless of numeric priority. This fixed two boundary failures identified in stress testing. MPAE is fully deterministic.

2.3 LLM Verifier and Model Configuration

When the kernel permits an action, Autex calls a configurable LLM to generate the natural-language response. The system prompt encodes all five policy rules as *unconditional constraints*, not guidelines. Temperature is fixed at 0 for Pass^k consistency. Extended thinking (Anthropic-shaped endpoints) or reasoning_effort (OpenAI-shaped endpoints) is forwarded to maximise reliability on boundary cases.

All configuration is environment-driven: AGENT_LLM, AGENT_API_KEY, AGENT_API_BASE, AGENT_TEMPERATURE, AGENT_REASONING_EFFORT, AGENT_THINKING_BUDGET_TOKENS, AGENT_API_STYLE. API shape (Anthropic Messages vs. OpenAI-compatible chat-completions) is auto-detected from base URL and model name, with an explicit override. Nothing is baked into the image.

2.4 Track 2: Sequential Call Structure

Track 2 allows at most 5 sequential LLM calls per baseline step and 500K average tokens per task. Autex uses two sequential steps:

Step	Kind	Purpose
1	Planner	Kernel + MPAE; zero LLM calls; classify action class
2	Verifier	Single LLM call: policy compliance + response generation
—	Parallel	Candidates within step 2 share sequentialStep=2; same depth

Token usage is reported via standard A2A turn_metrics only (prompt_tokens, completion_tokens, thinking_tokens). Sequential-call depth is documented here per Track 2 submission rules — not emitted as a custom metadata field. Three sequential steps remain in reserve for complex multi-turn tasks.

3 VALIDATION RESULTS

We ran 369 official CAR-bench trials across four frontier models using the benchmark's own harness:

Model	Avg P^3	Hall P^3	Dis P^3	P@3
Sonnet 4.6 (no think)	49%	46%	52%	79%
Opus 4.5 (thinking)	48%	48%	—	74%
Opus 4.6 (thinking)	47%	47%	—	72%
GPT-5.2 (think)*	59%	59%	—	81%

* GPT-5.2 tested on hallucination tasks only; not directly comparable.

arXiv:2601.22027.

Track: Track 2 (compute-use audit). Stress suite: 48/48 passing (final5 build).
Tasks: hidden split, all task types, 3 trials each.

Hallucination subtype breakdown:

Subtype	GPT-5.2	Op 4.6	Op 4.5	Snt 4.6
missing_tool	59%	46%	49%	48%
missing_parameter	100%	88%	75%	78%
missing_response	14%	14%	0%	6%

Hallucination_missing_tool_response is the dominant systemic weakness: 0–14.3% Pass^3 across all models. The missing_parameter subtype reaches 75–100% Pass^3, confirming that deterministic schema-level guards are highly effective. The 31.7pp gap (Pass@3 – Pass^3) for Sonnet 4.6 is the primary target of Autex's guard layer.

4 ABLATION AND LESSONS LEARNED

- **MPAE critical-context override:** without it, boundary cases (healthScore=59, temp=106°C) were misclassified as stabilize. One-line fix lifted MPAE to 11/11 stress-test cases.
- **Schema guards vs. runtime guards:** missing_parameter (schema-level) is highly effective; missing_response (runtime-level) requires LLM consistency to execute. Deterministic guards can only check what is visible before the LLM runs.
- **Model choice dominates system prompt quality:** original default (gpt-oss-120b, 28% Avg P^3) was the binding constraint, not the prompt. Switching to a provider-agnostic env interface removes the ceiling.
- **Temperature=0 reduces but doesn't eliminate variance:** boundary cases still diverge across trials. Hard unconditional rules in the system prompt reduce this further.
- **Stress suite as regression guard:** 48 deterministic cases caught the MPAE boundary bug and the metadata compliance violation (custom sequential-depth field) before submission.